

HOME NETWORK

Equipment & Infrastructure Documentation

This document provides detailed information on each piece of equipment within the home network, including hardware specifications, roles, configuration notes, and physical layout. It is intended as a reference for maintenance, troubleshooting, and future expansion.

1. Network Overview

The home network is organized across two physical enclosures: a 9U 10-inch switch rack that houses the core networking and compute equipment, and a three-tier Hyper Tough wire rack from Walmart that holds storage servers, additional compute nodes, and peripheral devices such as printers.

The network is logically segmented using VLANs to separate trusted devices (laptops, desktops, phones, consoles) from untrusted or smart home devices. Routing, firewall, and VLAN management are handled by an OPNsense installation running on repurposed enterprise hardware. DNS-level ad blocking is provided network-wide via Pi-hole running on a dedicated Raspberry Pi.

Network Topology Summary

Component	Details
Firewall / Router	Silver Peak FW-7551A-SV1 running OPNsense
Core Switch	GeeekPi 12-Port Cat6 Patch Panel + UGREEN 10-Port PoE Gigabit Switch
Trusted Wireless	TP-Link Omada Wi-Fi 7 WAP (main VLAN)
Untrusted Wireless	Aerohive AP330 (isolated VLAN for smart/legacy devices and guest network)
DNS Ad Blocking	Pi-hole on Raspberry Pi Zero 2 W
Print Server	Raspberry Pi 4B 8GB (color + B&W printers)
Hypervisor Cluster	Dell OptiPlex 7090 ×2 + Dell OptiPlex 3060 ×1 (Proxmox VE)
Network Storage	Dell OptiPlex 5040 running OpenMediaVault (OMV)

2. 9U 10-Inch Switch Rack

The 9U 10-inch rack is the central hub of the home network. It houses the firewall, patch panel, PoE switch, Raspberry Pi station, and the primary Proxmox cluster nodes. The compact 10-inch form factor makes it suitable for home use while still providing professional-grade cable management and expandability.

Position: Top (0U) — Firewall / Router (ThePillarOfAutumn)

Silver Peak FW-7551A-SV1 with OPNsense

Hardware	Silver Peak FW-7551A-SV1
Operating System	OPNsense (open-source FreeBSD-based firewall/router OS)
Acquisition	Purchased secondhand via eBay; OPNsense installed post-purchase
Form Factor	1U rackmount appliance (sits on top of the rack)
Role	Edge router, firewall, VLAN controller, DHCP server, and traffic manager

The Silver Peak FW-7551A-SV1 is a purpose-built network appliance originally designed for SD-WAN workloads in enterprise environments. In this deployment it has been repurposed as a home firewall and router by replacing the factory firmware with OPNsense. OPNsense provides a web-based management interface and supports advanced features including stateful packet inspection, VLAN segmentation, traffic shaping, DNS forwarding, and VPN (OpenVPN / WireGuard).

- **Primary function:** Manages all inter-VLAN routing and enforces firewall rules between trusted and untrusted segments.
- **DHCP:** Acts as the DHCP server for all VLANs, issuing leases and maintaining static assignments for key infrastructure devices.
- **DNS:** DNS requests are forwarded through Pi-hole (running on the Raspberry Pi Zero 2 W) for network-wide ad and tracker blocking.

1U — GeeekPi 12-Port Cat6 Patch Panel

Model	GeeekPi 12-Port Cat6 Keystone Patch Panel
Standard	Cat6 (supports up to 10 Gbps at short runs; 1 Gbps in typical home runs)
Role	Central patch and cable management point for all wired runs
Total Ports	12 (ports 1–12 with current assignments for ports 1–9 and 12)

The patch panel provides a single, organized termination point for all structured wiring in the home. Patch cables run from the panel to the downstream PoE switch, keeping cable runs neat and making future moves or changes straightforward.

Current port assignments:

- Ports 1–2: Wireless access points (one port per AP, runs to wireless access points)
- Ports 3–4: Raspberry Pi devices (print server and Pi-hole nodes in the Raspberry Pi station)
- Ports 5–6: Dell OptiPlex 7090 Proxmox nodes (one port per node for management and VM traffic)
- Ports 7–8: Dell OptiPlex 3060 and 5040 servers (Proxmox cluster node and OMV storage server)
- Port 9: Wired connection serving devices in the living room

- Ports 10–11: Reserved / unassigned (available for future use)
- Port 12: Uplink port to downstream switches or additional networking equipment

2U — UGREEN 10-Port PoE Gigabit Switch (TheArmory)

Model	UGREEN 10-Port PoE+ Gigabit Switch
Switching Capacity	Gigabit Ethernet (1000 Mbps) on all ports
PoE Standard	PoE+ (IEEE 802.3at) — provides up to 30W per port for PoE-powered devices
Role	Layer-2 switching and Power over Ethernet delivery to access points and other PoE devices

The UGREEN switch serves as the primary distribution switch in the rack. All patch panel connections terminate here, and PoE capability eliminates the need for separate power adapters for the wireless access points. Current connected devices include:

- TP-Link Omada Wi-Fi 7 access point (PoE-powered)
- Dell OptiPlex 7090 Proxmox nodes
- Dell OptiPlex 5040 OpenMediaVault storage server
- Raspberry Pi devices
- wired connections within home media center.

3U — Raspberry Pi Station

The Raspberry Pi station consolidates two small single-board computers into a single rack unit, each serving a dedicated infrastructure role.

Raspberry Pi 4B (8 GB) — Print Server (ForwardUntoDawn)

Hardware	Raspberry Pi 4 Model B, 8 GB RAM
OS	Raspberry Pi OS (64-bit) with CUPS (Common Unix Printing System)
Role	Network print server for both color and black-and-white printers
Connected Printers	Color laser/inkjet printer + monochrome laser printer (see Wire Rack section)

CUPS is configured to share both printers to all devices on the trusted VLAN. Client devices (Windows, macOS, Linux, iOS, Android) can discover and print to either printer without requiring locally installed drivers beyond the generic IPP/AirPrint support provided by the Pi. This centralizes print management and avoids the need to physically connect printers to individual devices.

Raspberry Pi Zero 2 W — Pi-hole DNS Sinkhole (SilentCartographer)

Hardware	Raspberry Pi Zero 2 W (quad-core ARM Cortex-A53, 512 MB RAM)
OS	Raspberry Pi OS Lite (headless)
Software	Pi-hole v5.x
Role	Network-wide DNS-based ad and tracker blocking

Pi-hole acts as the primary DNS resolver for all devices on the network. DNS queries are forwarded from the OPNsense firewall to the Pi-hole instance, which filters requests against community-maintained blocklists before forwarding legitimate queries upstream. This eliminates most display advertisements, telemetry endpoints, and known malicious domains across every device on the network—including phones, smart TVs, and consoles—without requiring per-device software.

4U–5U — Dell OptiPlex 7090 Proxmox Cluster Nodes (TheArkCluster)

Hardware	Dell OptiPlex 7090 Micro (quantity: 2)
Hypervisor	Proxmox Virtual Environment (VE)
Role	Primary Proxmox cluster nodes for containers (LXC) and virtual machines (KVM)
Cluster	Joined with Dell OptiPlex 3060 (located on wire rack) for a 3-node cluster

The two OptiPlex 7090 Micro units form the primary compute backbone of the Proxmox cluster. Proxmox VE is an open-source hypervisor platform based on Debian Linux that supports both KVM-based full virtual machines and lightweight LXC containers. The 7090 Micro's small footprint makes it well-suited for a 10-inch rack.

The Proxmox cluster (see also OptiPlex 3060 on the wire rack) enables live migration of VMs between nodes, high availability configurations, and centralized management through a single web UI. Typical workloads may include home automation servers, development environments, self-hosted services (media, productivity, monitoring), and networking utilities.

6U–8U — Cover Plates (Reserved)

Three rack units are currently covered with blank panels. These spaces are reserved for future expansion projects. Possible future uses include additional Proxmox nodes, a dedicated NAS expansion unit, a hardware firewall upgrade, or UPS (Uninterruptible Power Supply) integration.

9U — 8-Outlet Rackmount Power Distribution Unit (PDU) (Halo)

Type	8-outlet rackmount PDU with surge protection
Role	Powers all devices within the 9U switch rack; provides surge suppression

The PDU distributes AC power to all rack-mounted equipment and provides a first line of defense against power surges. All devices in the rack draw power from this unit. A future upgrade path would be to replace or supplement this PDU with a rackmount UPS to provide battery backup and runtime during brief power outages.

3. Wireless Access Points

Two access points provide wireless coverage throughout the home, each serving a distinct security zone. The segmentation ensures that untrusted or legacy devices cannot communicate with or interfere with trusted client devices.

AP 1 — TP-Link Omada Wi-Fi 7 Access Point (KeyshipRelay)

Model	TP-Link Omada Wi-Fi 7 (BE-series WAP)
Standard	IEEE 802.11be (Wi-Fi 7) — backward compatible with Wi-Fi 6/6E/5/4
Management	TP-Link Omada SDN controller (software-defined networking)
VLAN	Trusted main VLAN
PoE	Powered via PoE+ from UGREEN switch; no external power adapter required
Role	Primary wireless access point for all trusted client devices
Devices Served	Desktop computers, laptops, smartphones, tablets, video game consoles
SSIDs	2.4GHz: Installation04 / 5GHz: Installation05

Wi-Fi 7 (802.11be) introduces Multi-Link Operation (MLO), which allows devices to transmit and receive simultaneously across multiple frequency bands (2.4 GHz, 5 GHz, and 6 GHz) for lower latency and higher aggregate throughput. Even devices that do not support Wi-Fi 7 will connect at their highest supported standard (Wi-Fi 6 or 5) thanks to backward compatibility.

The Omada controller software provides centralized management, guest portal support, per-SSID VLAN tagging, QoS policies, and detailed client statistics. This AP is cabled back to patch panel port 1 or 2 and receives both data and power over a single Cat6 run.

AP 2 — Aerohive AP330 (GruntBarrier)

Model	Aerohive AP330
Standard	IEEE 802.11ac Wave 1 (Wi-Fi 5) — maximum link rate ~100 Mbps in this deployment
VLAN	Isolated untrusted VLAN (separate from trusted devices)
Role	Wireless access for untrusted, smart home, and legacy devices
Devices Served	Smart home devices (bulbs, plugs, sensors), original Xbox Live via Insignia service
SSID	GruntLobby (guest), XboxLive (used for Xbox)

The Aerohive AP330 is an older enterprise-grade access point repurposed for the untrusted device segment. All devices connecting to this AP are placed on a dedicated VLAN that is firewalled from the trusted network. Inter-VLAN traffic is blocked at the OPNsense firewall, so a compromised or misbehaving smart device cannot reach trusted hosts such as laptops or NAS drives.

Original Xbox consoles are supported through the Insignia service, a community-run replacement for the original Xbox Live infrastructure shut down by Microsoft in 2010. Because original Xbox titles use legacy network protocols and authentication servers, they are intentionally isolated on this AP alongside other legacy/untrusted traffic.

4. Wire Rack (Hyper Tough 3-Tier Shelving)

The wire rack is a three-tier Hyper Tough heavy-duty shelving unit purchased from Walmart. It serves as an open-frame equipment shelf for larger tower and desktop form-factor servers that do not fit in the 10-inch switch rack, as well as printers. The open-wire design provides natural airflow for equipment cooling.

Shelf 1 — OpenMediaVault Storage Server

Dell OptiPlex 5040 — OMV NAS (ONIVault)

Hardware	Dell OptiPlex 5040
Operating System	OpenMediaVault (OMV) — Debian-based NAS distribution
Role	Network-attached storage (NAS) for household files and media
Network	Wired Gigabit Ethernet via patch panel port 7 or 8

OpenMediaVault is an open-source NAS platform built on Debian Linux, offering a web-based administration interface for managing disk arrays, shared folders, user permissions, and network

protocols. The OptiPlex 5040 hosts the storage drives and makes files accessible to devices on the trusted VLAN.

Typical OMV services configured in this deployment may include:

- SMB/CIFS shares for Windows and macOS file access
- NFS shares for Linux clients and Proxmox VM disk images
- DLNA/media server for local media streaming to TVs and consoles
- Automated backup targets for workstations

The OptiPlex 5040 is a small desktop form factor machine, making it well-suited for repurposing as a low-power NAS. Storage capacity is determined by the drives installed in available SATA bays and any USB-attached expansion.

Shelf 1— Proxmox Cluster Node (TheArkCluster)

Dell OptiPlex 3060 — Third Proxmox Node

Hardware	Dell OptiPlex 3060
Hypervisor	Proxmox Virtual Environment (VE)
Role	Third node in the 3-node Proxmox cluster (with the two OptiPlex 7090s in the rack)
Network	Wired Gigabit Ethernet via patch panel port 7 or 8

The OptiPlex 3060 joins the two OptiPlex 7090 Micro units to form a three-node Proxmox cluster. A three-node cluster meets the minimum quorum requirement for high availability (HA) in Proxmox, meaning the cluster can tolerate the failure of a single node without losing the ability to manage or migrate virtual machines.

All three nodes share the same Proxmox cluster configuration and can be managed from a unified web interface. Storage for VM disk images can be local to each node or shared via the OMV NAS over NFS. The OptiPlex 3060 provides additional CPU and RAM resources for running containers and virtual machines that do not require the latest generation hardware.

Shelf 2-3 — Printers

Devices	Color printer + Black-and-white (monochrome) laser printer
Connectivity	USB → Raspberry Pi 4B Print Server (located in the switch rack)
Network Access	Available to all devices on the trusted VLAN via CUPS / IPP / AirPrint

Both printers connect via USB to the Raspberry Pi 4B print server in the Raspberry Pi station inside the switch rack. The print server runs CUPS, which exposes both printers as network printers over IPP (Internet Printing Protocol). Modern clients including Windows 10/11, macOS, iOS, and Android

can discover and use these printers automatically via Bonjour/mDNS without requiring manual driver installation.

- Color printer: For documents, photos, and graphics requiring color output.
- Monochrome laser printer: For high-volume black-and-white printing with lower per-page cost.

5. Future Expansion Notes

The following areas have been intentionally reserved or noted for future development:

- 6U–8U in the switch rack: Three blank cover plates reserve space for future rackmount equipment (additional nodes, UPS, patch panel expansion, etc.).
- Patch panel ports 10–11: Two Cat6 ports remain unassigned and are available for new wired drops.
- Power: The current 8-outlet PDU may approach capacity as additional equipment is added. A rackmount UPS with more outlets would add both capacity and battery backup.
- Storage: The OMV server's storage capacity can be expanded by adding drives to available SATA bays or attaching external USB storage enclosures.
- Proxmox cluster: The three-node cluster can be expanded with additional nodes. The current patch panel and switch can support at least two more wired devices before reaching port capacity.
- Wireless: A third access point could be deployed for a guest network or additional coverage in areas with weak signal using remaining patch panel ports.

6. Maintenance Reference

Recommended periodic maintenance tasks for the home network:

- OPNsense: Apply firmware and package updates monthly. Review firewall rules and DHCP leases quarterly.
- Proxmox: Update Proxmox VE and guest OS packages. Review VM/container resource allocation and backup schedules.
- Pi-hole: Update blocklists weekly (can be automated). Review blocked query logs periodically to identify false positives.
- OMV: Monitor disk health via SMART reports. Verify backup jobs are completing successfully. Check available storage capacity.
- Physical: Inspect cable routing bi-annually. Ensure rack ventilation is unobstructed. Check all patch cable connections are fully seated.
- Printers: Update CUPS and printer firmware as available. Clean print heads on color printer if ink-based.